Literature Review on Drug Development: Deucravacitinib (SOTYKTU)

A first-in-class oral allosteric TYK2 inhibitor for moderate-to-severe plaque psoriasis

Selected FDA-approved drug: Deucravacitinib (SOTYKTU), FDA approval: 9 September 2022

Abstract

Deucravacitinib, marketed as SOTYKTU, is an orally administered, selective allosteric inhibitor of tyrosine kinase 2 (TYK2). It received its first United States Food and Drug Administration (FDA) approval on 9 September 2022 for adults with moderate-to-severe plaque psoriasis who are candidates for systemic therapy or phototherapy. The drug represents an important development in immunopharmacology because it inhibits TYK2 through the regulatory pseudokinase domain rather than competing with ATP at the catalytic site, providing high selectivity over other Janus kinase (JAK) family members. This literature review summarizes the discovery and development pathway of deucravacitinib, its clinical trial program, therapeutic applications, and early market impact. Evidence from preclinical studies, first-in-human research, the phase II psoriasis program, and the pivotal phase III POETYK PSO-1 and PSO-2 trials demonstrates clinically meaningful efficacy with a convenient once-daily oral regimen. Long-term studies have also supported sustained efficacy and an acceptable safety profile. Commercially, SOTYKTU generated $170 million in worldwide revenue in 2023, $246 million in 2024, and $291 million in 2025, indicating continued uptake after launch. Overall, deucravacitinib illustrates how structure-guided drug discovery and selective pathway modulation can produce a clinically differentiated therapy in a competitive autoimmune-disease market.

1. Introduction

Psoriasis is a chronic immune-mediated inflammatory disease characterized by excessive keratinocyte proliferation and inflammatory signaling. Moderate-to-severe plaque psoriasis may substantially affect physical functioning, quality of life, and psychosocial well-being. Although biologic therapies can be highly effective, an important clinical need remains for oral, effective, and selective treatments with a mechanism that differs from conventional JAK inhibition. TYK2 is a member of the JAK family and participates in signaling downstream of cytokines including interleukin (IL)-12, IL-23, and type I interferons. Because these pathways are important in psoriasis, TYK2 became an attractive target for drug discovery.

Deucravacitinib was developed by Bristol Myers Squibb (BMS) as BMS-986165 and subsequently received the generic name deucravacitinib. It is distinguished from many earlier JAK inhibitors by binding the regulatory JH2 pseudokinase domain of TYK2. This allosteric mechanism stabilizes an inhibitory interaction between TYK2's regulatory and catalytic domains and prevents receptor-mediated activation. The FDA approved SOTYKTU in 2022, placing it within the five-year period specified for this assignment.

2. Drug Discovery and Development Process

The discovery program began with the identification of TYK2 as a therapeutically relevant kinase in autoimmune disease. A major challenge was achieving selectivity because the catalytic domains of JAK-family kinases are structurally related. BMS therefore pursued a different strategy: targeting the regulatory pseudokinase JH2 domain rather than the conserved ATP-binding catalytic site. Structure-guided design and optimization, including a water-displacement strategy, produced BMS-986165, a high-affinity JH2 ligand with strong TYK2 selectivity. Preclinical work demonstrated favorable pharmacokinetic properties and efficacy in animal models of autoimmune disease. The discovery work was reported as the development of a highly selective allosteric TYK2 inhibitor and the first pseudokinase-directed therapeutic to enter clinical development [1].

The next stage was translation into humans. The first-in-human study was a randomized, double-blind, single- and multiple-ascending-dose phase I study in 100 healthy volunteers. Deucravacitinib was rapidly absorbed, had an approximately 8–15-hour half-life, and showed dose-dependent pharmacodynamic effects on cytokine signaling. No serious adverse events were reported in the study, and the overall frequency of adverse events was similar between active treatment and placebo [2]. These findings supported progression to studies in patients.

In phase II, a randomized, double-blind, placebo-controlled study evaluated BMS-986165 in adults with moderate-to-severe psoriasis. At week 12, PASI 75 responses were substantially greater with several active doses than with placebo; for example, 69% of patients receiving 3 mg twice daily and 75% receiving 12 mg daily achieved PASI 75, compared with 7% receiving placebo [3]. These results established proof of concept for selective TYK2 inhibition in psoriasis and supported dose selection and further development.

The pivotal phase III program consisted primarily of POETYK PSO-1 and POETYK PSO-2. These were global, randomized, double-blind studies comparing once-daily deucravacitinib 6 mg with placebo and apremilast. POETYK PSO-1 enrolled 666 participants and POETYK PSO-2 enrolled 1,020 participants. At week 16 in PSO-1, 58.4% of patients receiving deucravacitinib achieved PASI 75 compared with 12.7% with placebo and 35.1% with apremilast. The sPGA 0/1 response was 53.6% with deucravacitinib, versus 7.2% with placebo and 32.1% with apremilast [4]. Across the two pivotal trials, the FDA's approval was based on evidence from 1,684 patients at 216 sites in 20 countries [5].

The regulatory milestone occurred on 9 September 2022, when the FDA approved SOTYKTU for adults with moderate-to-severe plaque psoriasis who are candidates for systemic therapy or phototherapy. The recommended dose is 6 mg orally once daily, with or without food [5,6]. The development pathway therefore followed the conventional sequence of target identification and medicinal chemistry, preclinical evaluation, phase I safety/PK studies, phase II proof-of-concept and dose-finding, phase III confirmatory trials, regulatory review, and post-approval studies.

3. Clinical Trial Phases

Phase I: The first-in-human study primarily evaluated safety, tolerability, pharmacokinetics, and pharmacodynamic effects. One hundred healthy volunteers received single or multiple ascending doses. The study demonstrated predictable absorption and biologic activity, supporting further clinical development [2].

Phase II: The phase II psoriasis study evaluated multiple oral doses and established proof of concept. The strong dose-related PASI responses showed that selective TYK2 inhibition could produce meaningful clinical improvement. The results also helped identify an effective regimen for phase III development [3].

Phase III: POETYK PSO-1 and POETYK PSO-2 were large, multinational, randomized, double-blind trials using placebo and apremilast as comparators. The main efficacy measures included PASI 75 and static Physician Global Assessment (sPGA) 0/1. Deucravacitinib demonstrated superiority to placebo and clinically important advantages over apremilast at key time points [4,7]. Patient-reported outcomes also improved, including psoriasis symptoms and signs and Dermatology Life Quality Index scores [7].

Long-term/post-approval evidence: Follow-up studies have examined sustained efficacy and safety. Two-year and three-year analyses reported continued clinical benefit, while a four-year analysis from the POETYK program provided additional evidence for long-term treatment. Such studies are important because psoriasis requires chronic therapy and regulatory approval alone does not replace ongoing safety surveillance [8,9,10].

4. Mechanism of Action and Therapeutic Applications

Deucravacitinib selectively inhibits TYK2 by binding its regulatory JH2 pseudokinase domain. This stabilizes an autoinhibitory interaction between the regulatory and catalytic domains and reduces downstream signaling. The mechanism is allosteric and differs from ATP-competitive inhibition of the JAK catalytic site. This molecular selectivity is a major feature of the drug's pharmacological profile [6].

The established FDA therapeutic application is moderate-to-severe plaque psoriasis in adults who are candidates for systemic therapy or phototherapy. It is administered as a 6 mg tablet once daily, which may be attractive for patients who prefer an oral treatment. Clinical trials showed improvements not only in lesion severity but also in patient-reported symptoms and quality of life [5,7].

Deucravacitinib has also been investigated for additional immune-mediated diseases. The development portfolio has included psoriatic arthritis, systemic lupus erythematosus, and other autoimmune indications. These are investigational applications unless and until supported by appropriate clinical evidence and regulatory approval. Thus, the distinction between approved and investigational indications is essential when describing the therapeutic scope of the drug [2,11].

5. Safety and Clinical Significance

The safety profile of deucravacitinib has been evaluated across randomized trials and longer-term extensions. Its selective mechanism was designed to avoid broader JAK-family inhibition, and clinical development has focused on characterizing infections, laboratory abnormalities, malignancy-related events, and other adverse outcomes relevant to immune-modulating therapy. In pooled phase III data, overall tolerability supported continued treatment through 52 weeks [10]. Long-term studies extending to three and four years have provided additional evidence for sustained efficacy and ongoing safety monitoring [8,9].

Clinically, the major significance of deucravacitinib is not simply that it is another systemic psoriasis medicine. It introduced a first-in-class allosteric TYK2 mechanism and offered a once-daily oral option with efficacy demonstrated against both placebo and an established oral comparator. This provides clinicians with another mechanism-based treatment choice within a disease area where patient preferences, previous treatment response, comorbidities, and long-term tolerability all influence therapy selection.

6. Market Impact

SOTYKTU entered the market in 2022 and has established a measurable commercial position within BMS's immunology portfolio. BMS reported worldwide SOTYKTU revenue of approximately $170 million in 2023, $246 million in 2024, and $291 million in 2025 [12,13]. This corresponds to continued year-over-year growth, although the product remains much smaller in revenue than BMS's largest medicines.

The market impact of deucravacitinib is broader than sales alone. First, it expanded the oral treatment segment for moderate-to-severe plaque psoriasis with a mechanism distinct from traditional oral JAK inhibitors. Second, it demonstrated the commercial and clinical potential of targeting the TYK2 regulatory domain. Third, its clinical-development program created a platform for evaluating TYK2 inhibition in additional immune-mediated diseases. BMS's continuing development portfolio lists SOTYKTU studies in conditions such as psoriatic arthritis, rheumatoid arthritis, systemic lupus erythematosus, and Sjögren's disease, showing that the company continues to explore the wider therapeutic potential of the molecule [11].

Competition remains substantial. Psoriasis treatment includes biologics targeting TNF, IL-17, IL-23 and other pathways, as well as established oral therapies. Therefore, continued market growth depends on comparative efficacy, safety, persistence, convenience, payer access, physician familiarity, and expansion into additional approved indications. Nevertheless, the rapid increase from $170 million in 2023 to $291 million in 2025 suggests meaningful commercial adoption after launch [12,13].

7. Conclusion

Deucravacitinib is a strong example of modern rational drug development. Its development began with a biologically validated target, followed by a deliberate strategy to overcome the selectivity limitations of conventional JAK inhibition. Structure-guided medicinal chemistry generated the allosteric TYK2 inhibitor BMS-986165, which progressed through preclinical studies, phase I human testing, phase II proof-of-concept studies, and large phase III POETYK trials. The evidence demonstrated clinically meaningful improvement in moderate-to-severe plaque psoriasis and supported FDA approval in September 2022.

The drug's first-in-class allosteric mechanism, once-daily oral administration, clinical efficacy, and expanding evidence base have made it an important addition to psoriasis pharmacotherapy. Long-term trials continue to characterize its benefit-risk profile, while additional studies are investigating its usefulness in other immune-mediated diseases. Commercially, increasing annual revenue demonstrates early market impact. Overall, deucravacitinib illustrates how target selectivity, translational pharmacology, rigorous clinical development, regulatory evaluation, and post-approval evidence can collectively transform a molecular concept into a clinically and commercially relevant medicine.

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